

COVARIANCE, CORRELATION AND SCATTER PLOTS

1. COVARIANCE

Covariance indicates whether two variables tend to vary in the same direction or in opposite directions. It describes the direction of the joint variation between two sets of observations.

- **Positive covariance:** Higher values of one variable tend to occur with higher values of the other.
- **Negative covariance:** Higher values of one variable tend to occur with lower values of the other.
- **Covariance close to zero:** No strong tendency to vary together in a linear fashion.
- The magnitude of covariance depends on the measurement scales of the variables, making it difficult to interpret strength solely based on the raw number.

$$\text{Cov}(X, Y) = \Sigma[(X_i - \mu_X)(Y_i - \mu_Y)] / N$$

Where:

- X_i = individual value of X
- Y_i = individual value of Y
- μ_X = mean of X
- μ_Y = mean of Y
- N = number of observations
- Σ = sum of the relevant values

The means are calculated as:

$$\mu_X = \Sigma X / N$$

$$\mu_Y = \Sigma Y / N$$

2. CORRELATION

Correlation is a standardized measure of the direction and strength of association between two variables.

The correlation coefficient (r) always falls between -1 and $+1$:

$$-1 \leq r \leq +1$$

- **+1** = perfect positive linear association
- **-1** = perfect negative linear association
- **0** = no linear correlation

Association vs. Causation:

Correlation describes association; correlation by itself does not establish causation. For example, you might observe a strong positive correlation between the sale of ice cream and the sale of sunglasses. However, buying sunglasses does not *cause* people to eat more ice cream. Instead, both are influenced by a third underlying factor: the weather (e.g., sunny, hot days). This clearly highlights the difference: two variables can move in relation to each other (association) without one causing the other (causation).

3. PEARSON CORRELATION

Pearson correlation measures the direction and strength of a linear association. Because it is standardized, it removes the scale dependency of covariance.

Once covariance is calculated, Pearson correlation is obtained by dividing covariance by the product of the standard deviations:

$$r = \text{Cov}(X, Y) / (\sigma_X \times \sigma_Y)$$

Since the standard deviations are the square roots of variances:

$$\sigma_X = \sqrt{\text{Variance}(X)}$$

$$\sigma_Y = \sqrt{\text{Variance}(Y)}$$

Therefore, the equivalent form is:

$$r = \text{Cov}(X, Y) / \sqrt{\text{Variance}(X) \times \text{Variance}(Y)}$$

PAGE 2 — COVARIANCE CALCULATION

Let us consider a realistic example tracking **Product Page Visits (X)** and the corresponding **Number of Orders (Y)** across 10 independent observation periods.

Observation	X (Visits)	Y (Orders)
1	5	6
2	10	4
3	12	8
4	15	6
5	18	11
6	20	9
7	24	14
8	28	10
9	32	17
10	36	15
Total (Σ)	ΣX = 200	ΣY = 100

STEP 1 — CALCULATE THE MEANS

Calculate the mean for both sets of values:

$$\mu_X = 200 / 10 = 20$$

$$\mu_Y = 100 / 10 = 10$$

STEP 2 — CALCULATION TABLE

Construct a detailed table to compute the deviations and cross-products. This single table provides all the necessary components for computing covariance, variances, standard deviations, and Pearson correlation.

X	Y	X - μ _X	Y - μ _Y	(X - μ _X)(Y - μ _Y)	(X - μ _X) ²	(Y - μ _Y) ²
5	6	-15	-4	60	225	16
10	4	-10	-6	60	100	36
12	8	-8	-2	16	64	4
15	6	-5	-4	20	25	16
18	11	-2	1	-2	4	1
20	9	0	-1	0	0	1
24	14	4	4	16	16	16
28	10	8	0	0	64	0
32	17	12	7	84	144	49
36	15	16	5	80	256	25
Σ=200	Σ=100	0	0	Σ = 334	Σ = 898	Σ = 164

STEP 3 — CALCULATE COVARIANCE

Using the sum of the cross-products from the table:

$$\text{Cov}(X, Y) = \Sigma[(X_i - \mu_X)(Y_i - \mu_Y)] / N$$

- 1. $\Sigma[(X_i - \mu_X)(Y_i - \mu_Y)] = 334$
- 2. $N = 10$
- 3. Substitution: $\text{Cov}(X, Y) = 334 / 10$
- 4. Division: 33.4

COVARIANCE = 33.4

Interpretation: The positive covariance indicates that higher values of X tend to occur alongside higher values of Y in the observations considered.

PAGE 3 — VARIANCE, STANDARD DEVIATION, PEARSON CORRELATION AND SCATTER PLOT

We continue with the same dataset of Product Page Visits (X) and Number of Orders (Y).

STEP 1 — CALCULATE VARIANCE OF X

$$\text{Variance}(X) = \Sigma(X_i - \mu_X)^2 / N$$

- $\Sigma(X_i - \mu_X)^2 = 898$ (from the table)
- Substitution: $\text{Variance}(X) = 898 / 10$

VARIANCE OF $X = 89.8$

STEP 2 — CALCULATE VARIANCE OF Y

$$\text{Variance}(Y) = \Sigma(Y_i - \mu_Y)^2 / N$$

- $\Sigma(Y_i - \mu_Y)^2 = 164$ (from the table)
- Substitution: $\text{Variance}(Y) = 164 / 10$

VARIANCE OF $Y = 16.4$

STEP 3 — CALCULATE STANDARD DEVIATIONS

$$\sigma_X = \sqrt{89.8} \approx 9.4763$$

STANDARD DEVIATION OF $X = 9.4763$

$$\sigma_Y = \sqrt{16.4} \approx 4.0497$$

STANDARD DEVIATION OF $Y = 4.0497$

STEP 4 — CALCULATE PEARSON CORRELATION

Using the values calculated in the previous steps:

$$r = \text{Cov}(X, Y) / (\sigma_X \times \sigma_Y)$$

$$r = 33.4 / (9.4763 \times 4.0497) = 33.4 / 38.3761 = 0.8703$$

Alternatively, using variances directly for precise calculation:

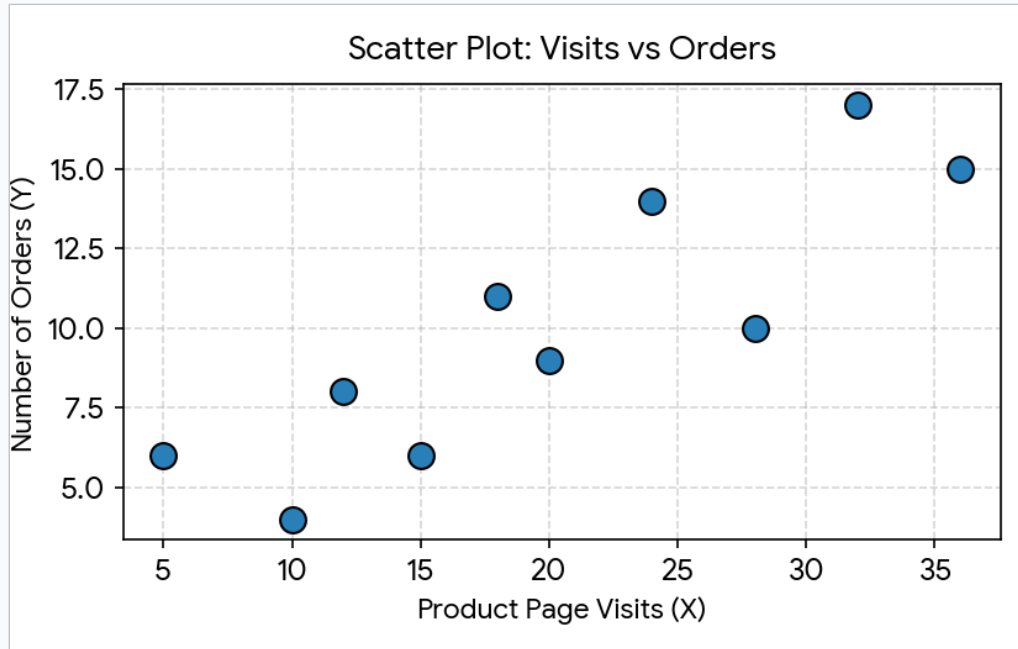
$$r = \text{Cov}(X, Y) / \sqrt{[\text{Variance}(X) \times \text{Variance}(Y)]}$$

$$r = 33.4 / \sqrt{[89.8 \times 16.4]} = 33.4 / 38.3760 = 0.8703$$

STEP 5 — INTERPRET PEARSON CORRELATION

Interpretation: The calculated correlation of 0.8703 indicates a strong positive linear association between the two variables. This means that as Product Page Visits increase, the Number of Orders generally increases in a proportional, linear manner.

STEP 6 — SCATTER PLOT



- The points generally travel from the bottom-left to the top-right, visually confirming the positive association.

PAGE 4 — SPEARMAN RANK CORRELATION

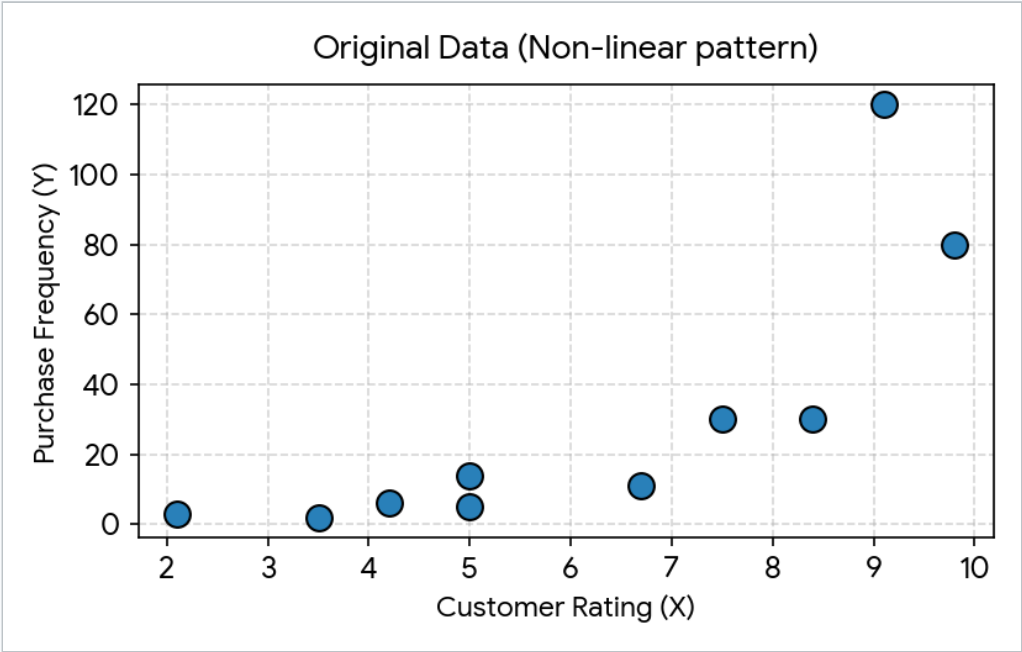
Spearman correlation evaluates the monotonic relationship between two variables using their ranks instead of their raw values. Let us look at a new dataset involving **Customer Rating (X)** and **Repeat-Purchase Frequency (Y)**. In this dataset, notice that some values are identical.

STEP 1 — ORIGINAL DATA

Observation	X (Customer Rating)	Y (Purchase Frequency)
1	2.1	3
2	3.5	2
3	4.2	6
4	5.0	5
5	5.0	14
6	6.7	11
7	7.5	30
8	8.4	30
9	9.1	120
10	9.8	80

ORIGINAL SCATTER PLOT & THE NEED FOR SPEARMAN

When we visualize the original data below, we see a distinct **non-linear (exponential) curve**. Because Pearson correlation only measures the strength of a *straight-line* relationship, it will underestimate the strength of this association. The Spearman Rank coefficient solves this by evaluating if the relationship consistently moves in the same direction (monotonicity), regardless of the curve's shape.



STEP 2 — ASSIGN RANKS (HANDLING TIES)

Convention: Rank 1 is assigned to the smallest value. When values are identical (tied), they receive the **average of the ranks** they would otherwise occupy.

- *X* has a tie at 5.0 (occupying ranks 4 and 5). Average rank = $(4 + 5) / 2 = \mathbf{4.5}$.
- *Y* has a tie at 30 (occupying ranks 7 and 8). Average rank = $(7 + 8) / 2 = \mathbf{7.5}$.

PAGE 5 — SPEARMAN CALCULATION: EXACT VS APPROXIMATION

The true mathematical definition of Spearman correlation is simply the **Pearson correlation applied to the rank variables**. Therefore, you must always find the ranks first. Let's calculate both methods based on the assigned ranks.

RANK CALCULATION TABLE

Let $d = \text{Rank } X - \text{Rank } Y$. The mean of Rank X (μ_{RX}) and Rank Y (μ_{RY}) are both 5.5.

Obs	Rank X	Rank Y	d	d^2	$(RX - \mu_{RX})(RY - \mu_{RY})$	$(RX - \mu_{RX})^2$	$(RY - \mu_{RY})^2$
1	1	2	-1	1	15.75	20.25	12.25
2	2	1	1	1	15.75	12.25	20.25
3	3	4	-1	1	3.75	6.25	2.25
4	4.5	3	1.5	2.25	2.5	1.0	6.25
5	4.5	6	-1.5	2.25	-0.5	1.0	0.25
6	6	5	1	1	-0.25	0.25	0.25
7	7	7.5	-0.5	0.25	3.0	2.25	4.0
8	8	7.5	0.5	0.25	5.0	6.25	4.0
9	9	10	-1	1	15.75	12.25	20.25
10	10	9	1	1	15.75	20.25	12.25
Σ	55	55	0	11	76.5	82.0	82.0

METHOD 1: THE EXACT METHOD (Pearson on Ranks)

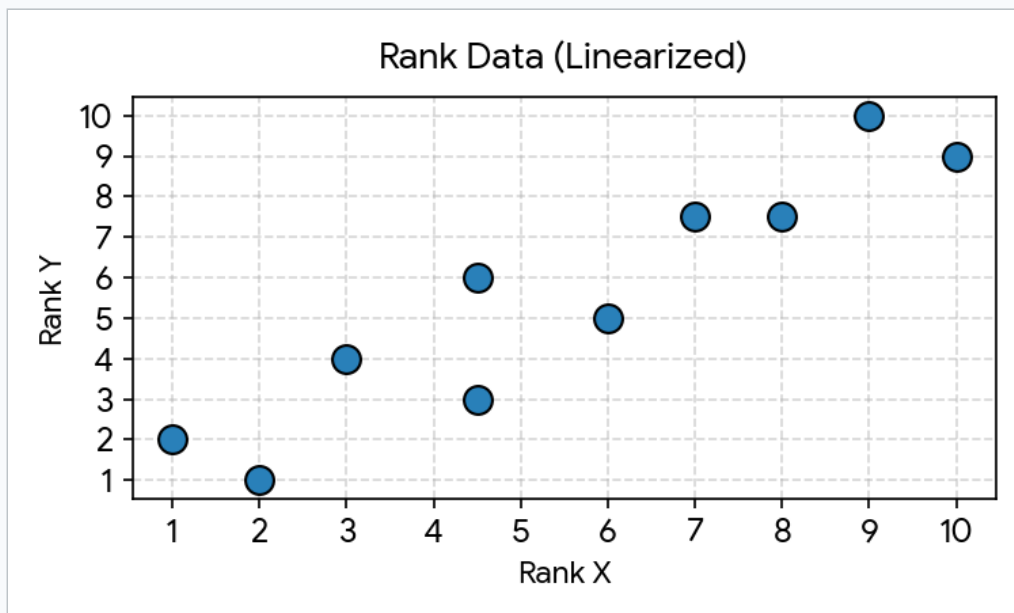
When ties exist, applying the standard Pearson formula to the ranks provides the exact Spearman correlation coefficient.

- $\text{Covariance}(RX, RY) = 76.5 / 10 = 7.65$
- $\text{Variance}(RX) = 82.0 / 10 = 8.20$
- $\text{Variance}(RY) = 82.0 / 10 = 8.20$
- $r = 7.65 / \sqrt{8.20 \times 8.20} = 7.65 / 8.20 = \mathbf{0.9329}$

METHOD 2: THE SHORTCUT APPROXIMATION

The famous $\rho = 1 - [6\Sigma d^2 / N(N^2 - 1)]$ formula is an algebraic simplification that assumes *no ties exist*. When ties are present, it is slightly inaccurate but serves as a close approximation.

- $\rho = 1 - [6(11) / 10(100 - 1)] = 1 - (66 / 990) = 1 - 0.0667 = \mathbf{0.9333}$



KEY TAKEAWAYS

- **COVARIANCE:** Indicates the direction of joint variation. A positive value means variables tend to move together; a negative value means variables tend to move in opposite directions. The magnitude depends on the measurement scale.
- **VARIANCE:** Measures the average squared deviation from the mean. It is used primarily to obtain standard deviation.
- **STANDARD DEVIATION:** Is the square root of variance. It expresses variability in the same measurement units as the original variable.
- **PEARSON CORRELATION:** A unit-free standardized measure ranging from -1 to $+1$. It measures linear association and can be calculated directly from covariance and standard deviations. It does not establish causation.
- **SPEARMAN CORRELATION:** Uses ranks and ranges from -1 to $+1$. It measures monotonic association and is useful when ordering or ranks are important. Tied ranks require appropriate treatment.
- **SCATTER PLOT:** Shows paired observations visually. It helps identify direction and assess strength, while revealing clusters, unusual observations, or possible nonlinear patterns.

COMPARISON TABLE

Feature	Covariance	Pearson Correlation	Spearman Correlation
Main purpose	Direction of joint variation	Direction and strength of linear association	Direction and strength of monotonic association
Based on	Original values	Original values	Ranks
Standardized	No	Yes	Yes
Range	Depends on measurement units	-1 to $+1$	-1 to $+1$
Unit-free	No	Yes	Yes
Relationship emphasized	Joint variation	Linear	Monotonic

FINAL FORMULA BOX

MEAN OF X

$$\mu_X = \Sigma X / N$$

MEAN OF Y

$$\mu_Y = \Sigma Y / N$$

COVARIANCE

$$\text{Cov}(X, Y) = \Sigma[(X_i - \mu_X)(Y_i - \mu_Y)] / N$$

VARIANCE OF X

$$\text{Variance}(X) = \Sigma(X_i - \mu_X)^2 / N$$

VARIANCE OF Y

$$\text{Variance}(Y) = \Sigma(Y_i - \mu_Y)^2 / N$$

STANDARD DEVIATION OF X

$$\sigma_X = \sqrt{[\text{Variance}(X)]}$$

STANDARD DEVIATION OF Y

$$\sigma_Y = \sqrt{[\text{Variance}(Y)]}$$

PEARSON CORRELATION

$$r = \text{Cov}(X, Y) / (\sigma_X \times \sigma_Y)$$

Equivalent form:

$$r = \text{Cov}(X, Y) / \sqrt{[\text{Variance}(X) \times \text{Variance}(Y)]}$$

SPEARMAN RANK CORRELATION

$$\rho = 1 - [6\Sigma d^2 / N(N^2 - 1)]$$

Note: The displayed Spearman formula applies to the no-ties case. When tied ranks occur, use an appropriate tie-aware calculation.